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PERFORMANCE STANDARDS FOR A UNIVERSAL
SHIPBORNE AUTOMATIC IDENTIFICATION SYSTEM (AIS)
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TABLE 1
Type of ship Reporting interval
Ship at anchor 3 minutes
Ship 0-14 knots 12 seconds
Ship 0-14 knots and changing course 4 seconds
Ship 14-23 knots 6 seconds
Ship 14-23 knots and changing course 2 seconds
Ship > 23 knots 3 seconds
Ship > 23 knots and changing course 2 seconds
Ship Reporting Capacity the system should be able to handle a minimum of 2,000 reports per minute to adequately provide for all operational scenarios envisioned.
6.3 Security
A security mechanism should be provided to detect disabling and to prevent unauthorized alteration of input or transmitted data. To protect against unauthorized dissemination of data, the IMO guidelines (Guidelines and Criteria for Ship Reporting Systems6) should be followed.

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the IMO guidelines (Guidelines and Criteria for Ship Reporting Systems6) should be followed.
A change to the IMO ship identification number in the equipment should only be possible by an authorized manufacturer's agent.
7 Permissible initialization period
The installation should be operational within 2 minutes of switching on.
8 Power supply
The AIS and associated sensors should be powered from the ship's main source of electrical energy. In addition, it should be possible to operate the AIS and associated sensors from an alternative source of electrical energy.
9 Technical characteristics
The technical characteristics of the AIS such as variable transmitter output power, operating frequencies (dedicated internationally and selected regionally), modulation and antenna system should comply with the appropriate ITU-R Recommendations.7
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6 Resolution MSC.433(98).
7 ITU-R M.1371 series refer.
RESOLUTION MSC.570(109) (adopted on 6 December 2024)

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RESOLUTION MSC.569(109) (adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY
INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION
BY MF AND HF DIGITAL NAVIGATIONAL
DATA (NAVDAT) SYSTEM
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ANNEX 12
RESOLUTION MSC.569(109)
(adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY
INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION
BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
RECALLING ALSO resolution A.886(21) on Procedure for the adoption of, and amendments to, performance standards and technical specifications, by which the Assembly resolved that the functions of adopting performance standards and technical specifications, as well as
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the functions of adopting performance standards and technical specifications, as well as amendments thereto, shall be performed by the Maritime Safety Committee on behalf of the Organization,
TAKING INTO ACCOUNT the amendments to the International Convention for the Safety of Life at Sea, 1974 ("the Convention"), adopted by resolution MSC.496(105),
NOTING, in particular, regulation IV/7.1.4 of the Convention, which requires every ship to be provided with a receiver or receivers capable of receiving maritime safety information and search and rescue related information throughout the entire voyage in w hich the ship is engaged,
NOTING ALSO the works carried out by the International Telecommunication Union on a digital navigational data (NAVDAT) system and the allocated frequencies on MF and HF for its use,
NOTING FURTHER the Guidance for the reception of maritime safety information and search
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use,
NOTING FURTHER the Guidance for the reception of maritime safety information and search and rescue related information as required in the Global Maritime Distress and Safety System (GMDSS) (MSC.1/Circ.1645), which identifies related information broadcast services and the equipment which should be installed on board ships to meet the requirements of chapter IV of the Convention,
RECOGNIZING that further growth in information promulgated to ships is constrained by the capacity of the International NAVTEX Service and HF NBDP broadcasts, and that MF and HF digital NAVDAT broadcasts may be used in the GMDSS,
HAVING CONSIDERED, at its 109th session, the recommendation made by the Sub-Committee on Navigation, Communications and Search and Rescue at its eleventh session,
1 ADOPTS the Performance standards for the reception of maritime safety information and search and rescue related information by MF and HF digital navigational data (NAVDAT)

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and search and rescue related information by MF and HF digital navigational data (NAVDAT) system, set out in the annex to the present resolution;
2 RECOMMENDS Governments to ensure that NAVDAT receiver equipment conforms to performance standards not inferior to those specified in the annex to the present resolution.
RESOLUTION MSC.569(109) (adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM

ANNEX
PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM
1 INTRODUCTION
1.1 The NAVDAT receiver equipment should be capable of receiving maritime safety information (MSI) and search and rescue (SAR) related information transmitted by NAVDAT

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information (MSI) and search and rescue (SAR) related information transmitted by NAVDAT system on MF and HF frequencies, and may be used to meet the requirements of SOLAS regulation IV/7.1.4.
1.2 In addition to meeting the requirements of the Radio Regulations, the NAVDAT receiver equipment should comply with the most recent versions of Recommendations ITU-R M.2010 and ITU-R M.2058 for NAVDAT in the MF and HF bands, respectively.
1.3 The NAVDAT receiver equipment should comply also with the requirements set out in resolutions A.694(17) , MSC.191(79), as amended by resolution MSC.466(101), and MSC.508(105), and the following performance standards.
2 GENERAL
2.1 The NAVDAT system allows the broadcast of messages in the form of digital files

providing texts, pictures or any other data. This broadcast can be done on the two international frequencies: 500 kHz in the MF band and 4 226 kHz in the HF band, and also on a ll other
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frequencies: 500 kHz in the MF band and 4 226 kHz in the HF band, and also on a ll other maritime radio frequencies assigned by the ITU for NAVDAT usage (see paragraph 4.2.1).
2.2 The NAVDAT receiver equipment should comprise a radio receiver, an appropriate antenna, a signal processor with non -volatile memory, a human -machine interface, data interfaces (see paragraph 9) and:
.1 an integrated display; or
.2 a connection to external equipment with a display (e.g. an integrated navigation system).
2.3 The NAVDAT receiver could either be a stand -alone equipment or combined with another equipment.
2.4 The NAVDAT receiver equipment should be provided with an antenna capable of receiving the entire maritime radio frequencies from, at minimum, 400 kHz to 30 MHz.
2.5 The NAVDAT receiver equipment should provide facilities to automatically update the ship's position and the time at which the position was determined from a suitable electronic
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ship's position and the time at which the position was determined from a suitable electronic position-fixing aid which may be an integral part of the equipment. For equipment which does not have an integral position -fixing aid, such facilities should include a suitable interface conforming to the appropriate international standards.1
2.6 When the NAVDAT receiver equipment has a dedicated antenna, it is recommended that the antenna be equipped with two outputs for sharing with another MF/HF receiver.
1 Refer to IEC 61162.

3 CONTROLS AND INDICATORS
3.1 Subject messages2 excluded by the operator from display should be readily available.